SAHIL BODKHE

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EDUCATION

Master of Engineering in Systems Design (AI & ML)	University of Waterloo at Ontario, Canada (4.0 GPA)	*Sep 2024- Aug 2025*
Bachelor of Engineering in AI & Data Science	University of Mumbai, India (9.55/10)	*Sep 2020 - May2024*
Honors Degree in Blockchain Technology	University of Mumbai, India	*Sep 2022 - May 2024*
Specialization in Business Studies	CONRAD SCHOOL OF ENTREPRENEURSHIP AND BUSINESS	Sep 2024- Aug 2025

Relevant courses: Algorithms & Data structures, Operating System, Web Development, Software Engineering & Project Management. AI, Machine learning, Industrial AI (Deployment of Deep Learning models), Systems Design, Recommendation Systems, Data Analytics, Database Management, Statistics, Big Data Analytics, Cloud Computing, Database Management, AI in Finance & Banking, Distributed Systems. Financial Accounting & Management, Leadership & Management.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Solidity

AI/ML: Scikit-learn, YOLOv5/v8, Optuna, Transformers, OpenAI SDK, RAG, Agents, LlAMA 3.2, Tensorflow, PyTorch , LORA, QLORA, PEFT .

Frameworks: FastAPI, Flask, Node.js, React.js, Spring Boot, LangChain .

Cloud & DevOps: AWS (EC2, S3), Docker, Postman, Git, MLFlow, DVC, Prometheus

Databases: PostgreSQL, MongoDB, ChromaDB, FAISS.

Tools: Ollama, MCP, Streamlit, Dagshub, Google Colab , Numpy , Pandas, MCP , Langgraph, LangSmith , A2A, ACP .

RECENT RESEARCH & PROJECTS

Agentic Banking Financial Assistant with LLM-Driven API Orchestration using Model Context Protocol (MCP) **May 2025 – July 2025**

- Developed a secure LLM-powered virtual assistant integrated with bank-grade backend APIs to handle wide range of customer queries right from simple balance checks, transaction history, to evaluating investment plans and projections.
- Enabled dynamic tool invocation using MCP (Model Context Protocol), ensuring contextual responses while preserving data privacy and regulatory compliance (e.g., PII handling, audit logging).
- The assistant intelligently handles user queries and dynamically routes them to backend APIs or generates query to access data if required in complex cases while maintaining zero exposure of sensitive data to public models.
- Achieved 80-90% reduction in support load by automating banking interactions, with a 95%+ accuracy in intent recognition and tool invocation. Focusing on delivering contextual, personalized, auditable conversations compliant with financial regulations
- Tech stack: Python, FastAPI, Llama3.2, Postman, StreamLit, ChromaDB, FAISS, LlamaIndex, Langchain , Langgraph , Langsmith .

Automated Tim Hortons Cup Detection System **Jan 2025 -April 2025**

- **End-to-End ML Pipeline Execution:** Spearheaded the complete machine learning lifecycle from data collection (4K+ custom/gen images), annotation, augmentation, and comprehensive analysis to model deployment with A/B testing, monitoring, and logging. Implemented a scalable Flask backend for seamless integration.
- **Advanced Model Optimization:** Fine-tuned YOLOv5/v8 architectures also used Optuna hyperparameter optimization, achieving 98.91% mAP50-95 (12% improvement over baseline) and significantly enhancing detection accuracy.
- **Bias Mitigation & Edge Case Handling:** Conducted in-depth dataset analysis, including t-SNE/UMAP clustering and stratified sampling, to diagnose and reduce bias, resulting in a 22% reduction in false positives and improved generalization to edge cases.
- **MLOps & Deployment:** Orchestrated the model lifecycle using MLFlow for experiment tracking, Dagshub, DVC data versioning, and CI/CD pipelines (Docker/AWS EC2) with Prometheus and custom API for real-time performance monitoring and logging.
- Tech Stack: Python, YOLOv5/v8, Optuna, MLFlow, Dagshub, DVC, Docker, Flask, AWS (EC2), Prometheus

Cricket Commentary Decoder for Automatic Database Updates. (Text Classification for Event Identification) ***Sep 2024-Dec 2025***

- Built NLP model on 80K commentary samples to classify cricket events (run, out, extras, dot) for API-triggered database updates.
- Evaluated models: LSTMs, GRU, Random Forest, SVM, XGBoost, AdaBoost.
- Enhanced event recognition by excluding all keywords (e.g., 'six', 'out', 'wide', 'four', 'byes' etc.) for robust context-based predictions.
- Boosted accuracy from 84% to 86% using LightGBM & Bi-LSTMs; also kernel SVMs achieved 84%+ in testing.
- Tech stack: Flask, Pandas, NumPy, Python, Scikit-Learn, Node JS, React JS.
- https://doi.org/10.21203/rs.3.rs-5712957/v1

Strategic Impact Recommender using Situational Analysis Predictions. *(Novel Strategy Forecast Application)* ***Sep2023 - May 2024***

- Developed sports research paper & web app for T20 cricket situational analysis and recommendation based on predictions.
- Invented Success Ratio Factor (SRF) formula to analyze game phases, enabling IPL teams to identify impact players with strategic

forecasts in every phase of the game.

- Cleaned, Transformed Normalized and Standardized 8 years of data (1.4M records) on MongoDB & Colab.
- Achieved 3/4 predictions correct in live match predictions during IPL 2024.
- Trained GradientBoostingRegressor with $R^2 = 0.8$, RMSE = 7; accurate score predictions at 15.3 and 20 overs.
- Built Compact User Interface & Rest API for fetching data from MongoDB and leveraging power of ML through Flask server.
- Tech stack: HTML, CSS, EJS, Node JS & Express JS, Flask, MongoDB Atlas, NumPy, Pandas, Scikit-learn, Python.

FarmEasy, Smart Inventory Management System for Farmers (Web App + ML in Agriculture Domain) **Mar 2024**

- Designed a Web application for managing, tracking, planning farm inventory and activities using dashboard.
- Implemented machine learning models for recommendations alongside with feasibility check integrating via Flask.
- Recommendation on quantity of farm tools, time requirement with respect to seeds, types of other logistics and land size.
- Recommendation of required tools to optimize resources for building new tools from existing ones.
- Designed the web features more user friendly by availing more visuals-oriented approach, translation, one click CRUD operations.
- Implemented chatbot for CRUD operations, website navigation, voice availability.
- Technology: HTML, CSS, React, NodeJS, ExpressJS, EJS, Flask, MongoDB, Postman, Google Colab.

Decentralized Freelance Talent Marketplace. *(Blockchain DApp in Finance Domain)* ***Nov 2023***

- Freelance platform with Escrow smart contract for fair payments & client-freelancer deal process.
- Creation of Auto payment release upon approval and dispute resolution with third-party arbiter or decentralized mechanism.
- Implemented smart contracts for milestone-based payment with different criteria like deadline, work Approval/Rejection, reason.
- Implemented of CRUD API for New Projects, Milestones, Profiles (Freelancers, Clients), Recommendations and bidding.
- Technology: HTML, CSS, JavaScript, Solidity, Hardhat.

INTERNSHIP & Experience

***Student Technical Intern*,**
TEXAS INSTRUMENTS & Ed Gate Technologies Pvt Lt, Bangalore. **February 2021.**

•Trained on TI microcontrollers and Software for designing diverse utilities.
•Gained hands-on experience in Embedded Systems and GUI development, enhanced technical and problem-solving skills.

CERTIFICATION & ACHIEVEMENTS

- AWS Solutions Architect.
- MongoDB M001 Certificate.
- 2 x Academic Excellence Award (Rank 1): (2022 & 2024)
- B.E Departmental All-Rounder (2024)
- 2 x Hackathons wins (Hackover 3.0 & Strategem)
- Highest Valuation Shark Tank CRCE 2023
- Built 15 + projects across different fields
- Participated in 15 + Hackathons.